Collection Revenge

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PROTOTYPE DOCUMENTATION

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Prototype Approach Learning Outcomes The prototype created for this project is a turn-based board game, the thought process behind this design is to simulate user choice moment by moment that would be happening in real-time in a video game. This extends to player choice in terms of how best to complete a game. Things that will be tested will be the player's choice when given a task and little direction on the best strategy to do so. Risks & Mitigations The first risk when testing fast-paced mechanics in a turn-based format is that players will have time to think about their options ahead of time instead of on the fly, the mitigation of this will be to implement a system where players will know their available options to them before playing.

The second risk is that in a turn-based game, there are mechanics that cannot be tested in this format, for example, things like the use of abilities in the way that translates to videogames cannot be done, one way to negate this is to adapt the abilities to a turn-based format by rewriting the concept.

Finally, the translation between the top-down perspective and FPS perspective creates a different feel for the game entirely in that players will have an advantage of seeing around them. To mitigate this, there will be elements of randomness that will deter the player from having additional advantages before the game starts. Playtest 1 Prototype Playtest Record https://www.youtube.com/watch?v=uHrAFmuOOGs Observations: Playtest Results The first initial physical test I initiated was to determine aspects of the map that would work best for an FPS perspective in relation to MOBA mechanics. primarily, this focused on testing the viability of sharp edges for turns as opposed to open spaced turns that allow the player to see more of the map. this extended to all areas on the map as the map was designed to be simple and easy to navigate. Next, this was used to test the advantages and disadvantages of vertical play. With an FPS perspective, players are able to utilize the sensation of looking up and down, with this comes map iterations that allow for more types of gameplay. mechanics like jumping would be used differently depending on what level of height you are at. These tests concluded that adding a level of height increases the number of interactions in terms of enemy AI that the player can experience. Adding verticality and open spaces allow for more area of the map to be seen, this coincides with my goal of combining FPS perspective and MOBA mechanics in that MOBA games are typically top-down where the player has the advantage of seeing all around them, however, in an FPS, players are limited to what they can see directly in front of them. Iterations This playtest has provided incite into both using spaces around corners and using height to enhance gameplay. for the prototype, future iterations will have an updated map design that showcases the finding from the playtest. although the findings on height-based perspective showed that verticality adds a chance for more mechanics, the findings do not translate well onto a turn-based physical copy so this data will be used in a future iteration of the video game format. Things to add will include different pathways that allow the user more player perspective. This change will allow for fewer moves to be made which will result in a different strategy to play the physical prototype of the game. In conclusion, this prototype highlighted more risk in translating an FPS into a turn-based game prototype, secondly, it provided incite into map design. With this in mind, future map iterations will have a unique map design that compliments the game style. Playtest 2 Prototype Playtest Record https://www.youtube.com/watch?v=_oDU7Ttf94s Observations: Playtest Results For this prototype test, I used

a game board to design a map for a turn-based game based on the findings in the last playtest. In this prototype, I wanted to research user strategy when faced with a task with no direction for the best way to go about it. This was done by creating a turn-based game where the objective was to move around the map with the use of dice, collecting nodes that were both located on the sides of the game map as well as on the main path of this map. Players must make their own decisions on which way to go as there are multiple solutions to the same puzzle, this study showed that in a turn-based setting, players were able to make moves towards the objectives with little error. This is due to the amount of time allowed for planning in each turn. Even when switching up the Node's order of collection, players were able to change up their strategy and find the most efficient route. Players quickly got bored if the same order was presented or if the nodes were not located in different spots. most games ended around 10 turns based on the D20 die used in this prototype. Iterations With the conclusion of this playtest, many iterations will be made both in regard to user interactions and their decision-making. The nodes in the prototype will now have a health value, this goes the same for the player pawn as well. with this, the player must now attack and eliminate the nodes to capture them. This change will force players to decide which nodes they should go for at the risk of their own hp. each node will require a certain amount of health to capture making it a puzzle for players to solve. another change will be to the die used for movement. for the next prototype, the use of a 6-sided die will be used to determine how many spaces the player can move. Doing this will slow the game down and allow for a more calculated approach when determining how the player should navigate the physical map. Playtest 3 Prototype Playtest Record https://youtu.be/0g4F9cWFiN0 Observations: Playtest Results This round of playtests included adding HP to both the enemy and the player character. The HP value of each node was predetermined based on how far away the node is from the player's initial spawn point. Contrary to past iterations of this prototype, the player would have to calculate the path they wanted to take to successfully capture all of the node points. After the first initial playtest of this version, we achieved our first loss in the game where the player did not have enough HP to complete the order. This led to a different strategy in future games where the player would calculate the least HP expensive path so the player could properly obtain all of the nodes. This strategy was also enabled by the use of the 6-sided die in that players had more time to recalculate their moves instead of taking multiple nodes in one turn. Players had a harder time completing the game with the new mechanics revolving around movement and Health. These tests served their purpose in creating user choice and slowing down the game. Our participants had to rapidly change their strategy from what it was in past iterations to successfully complete the objectives of the game. Iterations After analyzing the results of the last playtest, there are many variables that can be changed to increase the challenge aspect for the player in collecting the nodes on the map. Although adding HP values and the mechanic of "Attacking" forced the player to make up new strategies and different ways of playing, the game felt more like a puzzle in that the feeling of capturing nodes for the win was shadowed by the strategy that could be implemented before the game could begin. To mitigate this, in future iterations, the player will now have battle phases as a mechanic. After reaching each node, the player will go into a turn-based battle phase where they will roll two 6-sided dice. The value of the higher die roll will be used as the player's attack, whereas the die roll with the lowest value will be used for the node's attack. This will add a randomness effect to capturing nodes and deny the players an obvious pathway because they do not know what will happen in each battle phase. Playtest 4 Prototype Playtest Record https://youtu.be/ZBC939DqxeU Observations: Playtest Results Playtests this week added a mechanic of randomness to the game, doing this made the game feel less of a puzzle and more of an RPG-style adventure. Although the player used their initial strategy of determining the most convenient pathway before the game started, They were unable to stick with this strategy throughout the entirety of the game due to the fact that they would not have enough HP to successfully tank all the optimal nodes in their path. This was the desired outcome where the player was forced to make decisions in the game on the fly, contrary to their initial strategy developed at the start of the game. These changes in terms of user input worked well for the design of the research but quickly instated frustration in the player due to the randomness factor. In 2 of the playtests, the player simply gave up after realizing that their chances of winning were low based on how much health they had left. This outcome provided incite to how players were enjoying the game, unless it was clear that they had a chance to successfully collect all the nodes, they would enjoy winning, but the second that the game was determined "unwinnable" they lost interest in the completion of the game. Iterations The iterations of this prototype gave good feedback upon the research of user choice and decision in that they were forced to make decisions quickly every game to complete the goals. This simulated strategy is found in quick

decision-making FPS shooters and translated into a turn-based format. From a research standpoint, this worked well. Looking at this mechanic from the player's perspective, you can see that the randomness added more frustration to the game. To mitigate the frustration caused by randomness, in the next iteration, Health pickups will be added as a mechanic in the game. These health pickups are designed to alleviate the frustration when the player's health is low due to random rolls of dice. This will be achieved by increasing the strategy involved where the player may need to go out of their way to gain Health back. This adds another layer of user choice as the player will have to decide whether to capture a node or spend their turn gaining lost health back.

Playtest 5 Prototype Playtest Record https://youtu.be/eOYUzqfFa2c Observations: Playtest Results This week of iterations brought HP pickups to my game "Collection Revenge". This allowed players to recover HP on the map after battle phases were conducted. The HP spawn locations were randomized at the start of the game to ensure a different experience every time. The way the health pick-up works allowed the player to recover 10% of their HP for each pickup available, the strategy in this allowed for a lot of HP to be recovered if your health was low but yielded a small return when picked up at the beginning of the game. This was quickly picked up as the participant found that they could recover 2-3x more health if they waited till the end of the game (after they conducted most of the battles and collected almost all of the nodes). The strategy for completing the game changed drastically as the initial plan that was formulated by the player at the beginning of the game would always be modified by the randomness of damage caused by the nodes. The participant found it most fun when games seemed to be lost but could be saved if the player made calculated moves towards the health pickups. From there, they could remake their strategy with their new value of health and were able to beat the game. Iterations For these tests, giving the players a way to alleviate bad outcomes in the game proved to help with frustration as well as added user input to their strategy. The current HP system works on a percentage health mechanic that allows the pickups to scale based on the progression of the game. in theory, the player has a higher success at capturing nodes if they hold out on using health pick up till the last 50% of nodes are available. Although this is great for the player, the battle phases in the game lose meaning when the player understands that they can win and gain all of the HP back after the Battle. In future iterations of this prototype, the mechanic of gaining HP back will be designated to an ability usage. Once per battle phase, the player can activate a "Leech life" ability where the player is able to roll two 6-sided dice for their attack value, the player then deals the damage of the two die rolls and then recovers HP based on the damage done (12 damage = 12 HP gained). This will add strategy to the player's battle phase in that they must use their abilities to recover hp. This will replace the current HP pick-up mechanic and allow the player to recover HP while completing objectives.

Playtest 6 Prototype Playtest Record https://youtu.be/VcQ1PzlyGvY Observations: Playtest Results With the addition of HP recovery, my game "Collection Revenge" feels more like an RPG now that the character has abilities that they can choose to use. The new "leech life" ability was used consistently for every battle phase in this prototype which allowed the user to gain HP back 4 times instead of 5 (per the last iteration where there were 5 HP pickups on the map in total). This averaged out where the player gained less HP in total per game but was still able to successfully collect the game nodes and complete the game. In multiple of the tests, the participant got close to death with one game resulting in a win with a remaining HP value of 1. This game sparked the most excitement in almost failing but still coming out on top. This mechanical change allowed the user more user input where they can decide the most optimal time to use their ability but still relied on randomness for each ability value in terms of HP gained back. If the participant chose to use their ability and rolled a small sum (in some cases, gaining 2 HP) this would lead to feeling like they did not really get to use their ability to its potential because it was a one-time use mechanic. Iterations This round of Prototype testing was used to test HP mechanics in terms of user input and the feeling that the player feels when they feel like they are the reason that they won the game. These tests on player abilities showed that the participants enjoyed being able to use their own skills to recover their HP when the rolls of the ability were high (EX: 7-12) but when receiving a low sum (EX:1-6), the participant stated that they did not feel like they had control over their own abilities. in future iterations of this game, to mitigate the feeling of an unsuccessful ability attack, the player will now be given the chance to roll 2 more additional times, the first reroll of the two 6-sided dice allows you to choose between the first 2 rolls for their attack value, if the player is unhappy about

their first 2 rolls, the player will also be given the chance to roll one more additional time in a last-ditch effort to gain a better roll.

Playtest 7 Prototype Playtest Record https://youtu.be/QerFpCFOpqE Observations: Playtest Results "Collection Revenge" had the most success in terms of user experience for this round of prototype testing in that players were able to have more in-depth control of the battle phase. The implementation of a reroll system into character abilities allowed players to have a positive outcome when using the "leech life" ability. Participants stated that it felt more like the previous HP pickup system in terms of difficulty, where players would recover HP by collecting the pickups. This made the game more predictable as the player was able to always have high damaging abilities and HP recovery. In multiple games, the player used the reroll system and got a low score but felt that it was fair because it was their choice to go for the "all or nothing" roll. From analyzing the gameplay, the general difficulty of the game has gone down, players feel like they can successfully collect all of the nodes and make it to the end. Participants seemed to enjoy this iteration the best as they were able to decide choices on each turn and change their strategy to better fit their gameplay. Finally, I hope to relate these findings to a video game format where players are forced to make choices quickly and adapt to what is happening. Iterations These playtests proved to provide players with even more user choice, this increased the amount of enjoyment the players had in that the times where the player struggled felt fair because they were in the position based on the choices they picked. Although players had an easier time utilizing their abilities, the difficulty of the game was lowered due to the reroll system. in future iterations of the game, I would like to add an additional mechanic for the nodes where they are able to use an ability every 3rd turn. This will increase the damage output of the enemies in the game making it harder for the player to win. They will have to quickly choose which nodes to go after and calculate how many turns it will take. This research will provide insight into when the player uses their abilities and if they develop a strategy based on the changes to the enemies.

Playtest 8 Prototype Playtest Record https://youtu.be/gBDWhmk4af8 Observations: Playtest Results The prototype for this iteration of "Collection Revenge" improved upon the enemies' interactions in the battle phase. A new mechanic was added to do this in that every 3rd turn in the battle phase, the nodes would make an additional attack. This attack is based on the value of a dice roll which then gets halved, giving you the damage for the additional attack. This change was used to enhance player strategy in the timing of their abilities. When playing this iteration of the game, players felt frustration with the first few attempts all of which resulted in a close win. Participants stated that this new change affected how they would use abilities in the game where the best optimal use was to use it after the enemy had used their ability. The new mechanic in the game changed how players would make decisions in the battle phases of the game, while simultaneously adding an additional layer of difficulty when playing the game. After multiple games, a strategy was determined but players noted that the game felt "Stale" after playing it over with their newfound strategies. This prototype provided useful information regarding replayability in the game and how mechanics may make the game feel more scripted. Iterations This iteration for the game "Collection Revenge" helped better understand player strategy in that players were able to adapt to game changes and use their abilities at optimal times. Players enjoyed the new changes to the enemy nodes for the first 2 times playing it through but once they were able to master the mechanic, the game became easy again. In future iterations of the game, I would like to implement a blocking system that allows the player to block specific turns of attack for the enemy. This will be used to increase the player's user input and strategy where they will be tasked with finding the best opportunities to use this new ability. This ability will be gained by picking up a "Block" node found around the map. This system will increase map exploration and increase the amount of strategy needed to complete the main objective of collecting all of the nodes.

Conclusions The conclusion of the prototype playtest shows a clear correlation between the player input and fun. I learned that making changes that affect the randomness of the game can leave a bitter taste in players' mouths if not paired properly with a way to alleviate frustration with randomness. Making changes that allow for more user input and gets players thinking on their feet will increase the amount of fun in each turn. This correlated to the research initially examined in the second prototype where the design was to allow for players to have a choice in their decisions. in doing this, I was able to gather data regarding the choice players

made between pathways to choose, nodes to collect, abilities to use, and finally, the optimal time to use said abilities. To finalize these findings, players had to play through many iterations of design where the moves they made and strategies they discovered would be used to enhance the next iteration of the game. this method allowed me to quickly make changes that increased aspects of the game including the addition of a battle phase and a more RPG feel to the turn-based game. The information gathered in this research will be used in future iterations of games I work on and change how user input is determined. Original Rules Prototype Rules V1

RED - Main Nodes (MN) GREEN - Camp Nodes (CN) BLUE - Structure Nodes (SN) GREEN (Stripped)- Player start

Items: D20 die *D6 die for iterations, 2nd prototype* Game board: *added Health pickups, prototype 4 *added Health values for enemies, Prototype 2

Setup: *added, prototype 2 Roll a d20 die to determine the hp value of each node by rolling once and recoding the number for every node. The player starts with 100 HP HP pick ups are placed at random based on where the 5 game pieces land when dropped on the board - Prototype 3

Rules: The player is tasked with collecting nodes in a specific order.

The player can start on any of the available Green-striped spaces. Your goal is to collect all of the nodes on the map in a rhythmic order in as little of turns as possible. Order: Camp Node (CN) (ability every 3rd turn - Prototype 7) Structure Node (SN) Camp Node (CN) (ability every 3rd turn - Prototype 7) Structure Node (SN) Either Main Node (MN) (ability every 3rd turn - Prototype 7) Players can move once per turn using a d20 die to determine the value of the roll (EX: D20 roll = 15 - Move 15 spaces) After moving to an available node, the player collects it and adds it to their inventory. [(The player takes damage equal to the nodes HP) - Prototype 2] [(The player then goes into the battle phase) - Prototype 3] The player will take turns against the node to capture it, roll a 6d for the players attack and then roll again to simulate the nodes attack. - Prototype 3 once in each battle phase, players can choose to use an ability to gain HP back, Roll two 6D die, and subtract that number from the Nodes HP while also adding the value to your own HP. - Prototype 4 In each battle phase, when using an ability, roll twice to and select the higher number for your ability attack, if neither option is optimal, you may roll one more additional time but you must go with that value. - Prototype 6 When the player has collected all of the required camp and structure nodes, they make capture either of the Main Nodes and win the game.

Final Rules Prototype Rules (Final)

RED - Main Nodes (MN) - use abilities every 3rd turn GREEN - Camp Nodes (CN) - use abilities every 3rd turn BLUE - Structure Nodes (SN) GREEN (Stripped)- Player start

Items: D6 die (2 optimal) D20 die Game board:

Setup: Roll a d6 die to determine the hp value of each node by rolling once and record the number for every node. The player starts with 20 HP

Rules: The player is tasked with collecting nodes in a specific order.

The player can start on any of the available Green-striped spaces. Your goal is to collect all of the nodes on the map in a rhythmic order in as little of turns as possible. Order: Camp Node (CN) (ability every 3rd turn) Structure Node (SN) Camp Node (CN) (ability every 3rd turn) Structure Node (SN) Either Main Node (MN) (ability every 3rd turn) Players can move once per turn using a d6 die to determine the value of the roll (EX: D6 roll = 6 - Move 6 spaces) After moving to an available node, the player initiates the Battle phase. If the player was won the battle phase, collect the node. When the player has collected all of the required camp and structure nodes, they make capture either of the Main Nodes and win the game.

Battle phase: The player always goes first Take turns rolling a die to determine damage values for you and the enemy node. The battle ends when either the player HP = 0 or the nodes HP = 0 The player may use their ability (Leech Life) once per battle phase to attack the opponent, to do so, roll a pair of 6d dice twice and record the value for each, then the player has the option of either value or rolling one additional time but they must play the last roll. After damaging the opponent, gain the HP they lost as HP for your player.